csiro-biomass

9th Place Solution

Rank	9
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Version	Original

Source: https://www.kaggle.com/competitions/csiro-biomass/discussion/671026
Retrieved with Kaggle CLI on 2026-07-18.

Our solution is a weighted ensemble of **three model variants**, all built on the DINOv3 ViT-Huge backbone with LocalMamba fusion blocks. The key thing that helped was addressing the challenging `Dry_Dead_g` target through ratio-based prediction and multi-head architecture variations.

Final Ensemble: 3 model architectures × 4 folds.

Model Architecture

Shared Components

All models share the following core architecture:

```
Input: Image (2000×1000) → Split into Left (1000×1000) + Right (1000×1000)
    ↓
DINOv3 ViT-Huge Backbone (vit_huge_plus_patch16_dinov3.lvd1689m)
    - Pretrained weights from timm
    - Gradient checkpointing enabled for memory efficiency
    - global_pool='' to retain patch tokens
    ↓
Feature Concatenation: [Left Tokens, Right Tokens] → (B, 2N, D)
    ↓
LocalMamba Fusion (2 blocks)
    - LayerNorm → Gated Linear → DepthwiseConv1d → Projection
    - Residual connections
    ↓
Adaptive Average Pooling → (B, D)
    ↓
Multi-Head Regression with Softplus activation (non-negative outputs)
```

Model 1: Improved Dead Prediction (Ratio-Based)

Key Innovation: Predict `Dry_Dead_g` as a ratio of `Dry_Total_g`

Head	Output	Activation
`head_green`	Dry_Green_g	Softplus
`head_clover`	Dry_Clover_g	Softplus
`head_total`	Dry_Total_g	Softplus
`head_dead_ratio`	Dead/Total ratio	**Sigmoid** [0,1]
Derivation:		

- `Dry_Dead_g = head_dead_ratio × Dry_Total_g`
- `GDM_g = Dry_Green_g + Dry_Clover_g` **Special Handling:**
- State-specific Dead adjustment via learnable bias (WA state has Dead≈0)
- Multi-task loss with auxiliary ratio loss

- Scale-invariant log-space loss for Dead **Training Config:**
- Image Size: 1024×1024
- Loss Weights: [0.12, **0.25**, 0.12, 0.16, 0.35] (increased Dead weight)

Model 2: Standard Multi-Head

Architecture: Direct prediction with derived targets

Head	Predicted	Derived
`head_green`	Dry_Green_g	-
`head_dead`	Dry_Dead_g	-
`head_clover`	Dry_Clover_g	-
-	GDM_g	Green + Clover
-	Dry_Total_g	GDM + Dead
Training Config:		

- Image Size: 1024×1024
- Loss Weights: [0.15, 0.15, 0.15, 0.2, 0.35]

Model 3: Total-First Prediction

Key Difference: Predict Total and GDM directly, derive Dead

Head	Predicted	Derived
`head_green`	Dry_Green_g	-
`head_clover`	Dry_Clover_g	-
`head_gdm`	GDM_g	-
`head_total`	Dry_Total_g	-
-	Dry_Dead_g	**Total - GDM**
Training Config:		

- Image Size: 896×896
- Loss Weights: [0.15, 0.15, 0.15, 0.2, 0.35]

Training Strategy

Data Preprocessing

Image Split: Panoramic images (2000×1000) split into Left half (0:mid) and Right half (mid:end), processed separately through the backbone.

Cross-Validation Strategy

The CV Strategy we used was suggested by someone in the discussion. I am not able to find that thread, but thanks to him. we used **StratifiedGroupKFold** (4 folds) with a carefully designed stratification and grouping scheme: **Stratification Label:** Based on presence/absence of challenging targets

```
stratify_col = f"{has_clover}_{has_dead}"
# Ensures each fold has balanced representation of:
#   - Samples WITH Clover (Dry_Clover_g > 0)
#   - Samples WITH Dead (Dry_Dead_g > 0)
```

Group Key: Prevents data leakage by grouping samples from same collection session

```
group_key = f"{day}/{month}_{state}"
# Examples: "15/3_NSW", "22/6_WA", "8/11_Vic"
# Ensures samples collected on the same date in the same state stay together
```

Why This Matters:

1. **Stratification by Clover/Dead:** `Dry_Clover_g` is often zero (sparse), and `Dry_Dead_g` has high variance. Stratifying ensures each fold sees similar distributions of these challenging targets.
2. **Grouping by Date+State:** Samples collected on the same day in the same location likely have similar conditions (weather, pasture state). Keeping them in the same fold prevents the model from "memorizing" collection-day patterns.

```
sgkf = StratifiedGroupKFold(n_splits=4, shuffle=True, random_state=42)
for fold, (_, val_idx) in enumerate(
    sgkf.split(df, df['stratify_col'], groups=df['group_key'])
):
    df.loc[val_idx, 'fold'] = fold
```

Augmentations (Training)

```
A.Resize(IMG_SIZE, IMG_SIZE)
A.HorizontalFlip(p=0.5)
A.VerticalFlip(p=0.5)
A.RandomRotate90(p=0.5)
A.Rotate(limit=(-15, 15), p=0.3, border_mode=REFLECT)
A.ColorJitter(brightness=0.15, contrast=0.15, saturation=0.15, hue=0.05, p=0.5)
A.GaussNoise(var_limit=(5.0, 20.0), p=0.2)
A.Normalize(ImageNet stats)
```

Hyperparameters

Parameter	Value
Batch Size	2
Gradient Accumulation	8 (effective batch = 16)
Epochs	100 (early stopping)
Freeze Epochs	10
Warmup Epochs	3
LR Backbone	1e-5
LR Head	1e-4
Weight Decay	1e-2
EMA Decay	0.998
Patience	15
Gradient Clipping	1.0
Mixed Precision	FP16

Loss Function

Weighted Huber Loss (β=5.0):

```
loss = Σ(weight_i × SmoothL1Loss(pred_i, target_i, beta=5.0))
```

Model 1 Additional Losses:

- Auxiliary MSE loss on Dead/Total ratio (weight=0.15)
- Scale-invariant log-space loss for Dead (weight=0.05)

Inference & Ensemble

Validation TTA

4-way TTA during validation:

1. Original
2. Horizontal Flip
3. Vertical Flip
4. Horizontal + Vertical Flip

Ensemble Strategy

Weighted Average with Fold-Specific Weights:

```
weight = {
    fold_0: 1.25,
    fold_1: 0.75,
    fold_2: 1.25,
    fold_3: 0.75
}
ensemble = Σ(weight × fold_predictions) / Σ(weights)
```

Post-Processing

```
if target_name == "Dry_Clover_g":
    prediction *= 0.8
elif target_name == "Dry_Dead_g":
    if prediction > 20:
        prediction *= 1.1
    elif prediction < 10:
        prediction *= 0.9
```

Key Insights

1. **Dead Prediction is Hard:** `Dry_Dead_g` has the highest variance and lowest correlation with other features. Ratio-based prediction (Dead/Total) significantly improved stability.
2. **State Matters for Dead:** WA samples consistently have Dead=0. Learnable state-specific bias helped the model learn this pattern.
3. **Resolution Trade-off:** 1024×1024 performed slightly better but 896×896 was faster; combining both helped. In fact, for the models trained on 896, inferring them on 1024 performed better than inferring them on 896.

Results

Model / Ensemble	Public Score	Private Score
Model 1: Ratio-Based Dead	0.76760	0.65329
Model 2: Standard Multi-Head	**0.77157**	0.64714
Model 3: 4-Heads (Total-First)	0.76685	0.64803
Ensemble (with fold weights + post-processing)	0.77142	**0.65179**
Ensemble (no fold weights, no post-processing)	0.76772	0.64855

What Didn't Work

- Metadata conditioning (State/Season embeddings) did not work
- We did not use TTA for the submission of ensemble as it was exceeding 9 hrs.
- Higher image resolutions (>1024)
- More Mamba blocks (>2)